

Quiz 3 2026
Biophysics(BIO215)

Time: 45 minutes

Total Marks:20

Section-A (MCQ 1*10 Marks)

Q1. The posre.itp file generated by pdb2gmx contains position restraints. For which atoms does it apply restraints?

- A. All atoms including hydrogens that were added by pdb2gmx.
- B. Only the alpha carbon backbone atoms.
- C. Heavy atoms (non hydrogen) from the original PDB file.
- D. Only water oxygen atoms in the solvated system.

Q2. What key difference distinguishes the NPT equilibration from NVT equilibration in the GROMACS workflow?

- A. NPT uses a verlet integrator while NVT uses leap frog.
- B. NPT allows the simulation box to change size via pressure coupling.
- C. NVT uses position restraints while NPT has no restraints at all.
- D. NPT generates new velocities from a Maxwell distribution for each step.

Q3. Which analysis gives the per-residue flexibility of a protein after an MD simulation and can be visualized by coloring a PDB structure by B-factor in PyMOL or Chimera?

- A. RMSD (Root Mean Square Deviation).
- B. RMSF (Root Mean Square Fluctuation).
- C. Radius of gyration.
- D. H-bond analysis.

Q4. To define the docking grid box, a researcher identifies four active site residues of receptor MLK1 with the following last-atom coordinates:

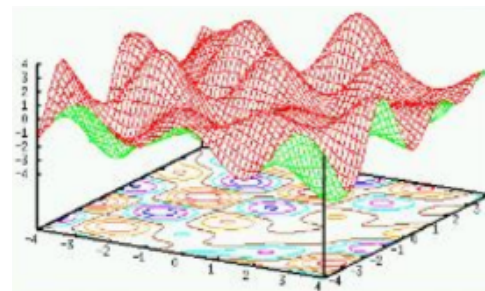
GLU221 → (25.945, 45.067, 22.98) | ALA223 → (25.367, 38.703, 26.074) | ARG230 → (22.937, 31.658, 32.491) | SER272 → (28.59, 38.55, 38.448)

What are the approximate X, Y, Z coordinates of the grid box centre?

- A. (25.71, 38.34, 30.00)
- B. (28.59, 45.07, 38.45)
- C. (22.94, 31.66, 22.98)
- D. (51.42, 76.67, 60.00)

Q5. In an energy landscape diagram showing multiple minima, which of the following best describes why a system preferentially settles into a deep, narrow well?

- A. Deep wells have higher energy, making them thermodynamically unstable
- B. The system is more likely to be found in states of high potential energy



- C. A deep well represents a low-energy state that the Boltzmann distribution heavily favors
- D. The width of the well has no relation to the number of available states

Q6. A chemist writes the ideal gas law as $pV = nRT$, while a physicist writes it as $pV = NkT$. What is the correct relationship between R and k ?

- A. $R = k/N_a$
- B. $R = k \cdot N_a$
- C. $R = k^2$
- D. $R = N_a/k$

Q7. Coulombtype = PME implies:

- A. Only short-range electrostatics are computed
- B. Electrostatics are truncated at cutoff
- C. Electrostatics are split into real and reciprocal space
- D. Electrostatics are ignored

Q8. Why is energy minimization performed before any MD simulation, even though the structure may already look correct from crystallography?

- A. To assign velocities before the simulation starts.
- B. To set the temperature of the system to 300K and mimicking it to normal temperature conditions.
- C. To relieve steric clashes and bad geometry introduced during structure preparation (e.g., added hydrogens), preventing simulation crashes due to abnormally large forces.
- D. To generate the .tpr binary run file required by mdrun to proceed to next step of equilibration.

Q9. In an MD simulation using Periodic Boundary Conditions (PBC), what happens when an atom reaches the edge of the simulation box?

- A. It bounces back elastically like a billiard ball
- B. The simulation pauses and the atom is repositioned manually.
- C. It re-enters from the opposite face of the box, mimicking a bulk environment.
- D. It is deleted and a new atom is created at the center.

Q10. When a protein structure is downloaded from RCSB PDB, it contains crystallographic water molecules inside and around the binding pocket. Why must these be removed before docking?

- A. Water molecules increase the file size, slowing down the software unnecessarily
- B. Water molecules occupy the binding cavity and would block the ligand from being placed in the active site during pose generation
- C. AutoDock Vina cannot read oxygen atoms from water molecules
- D. Water molecules make the protein positively charged, disrupting Kollman charge assignment

Section B(2 mark each)

Q1. Ludwig Boltzmann and James Clerk Maxwell both contributed to the statistical description of gases. Briefly distinguish the specific contribution of each to our understanding of molecular distributions, and explain what the quantity kT physically represents in the Boltzmann distribution.

Ans: Maxwell derived the speed distribution of gas molecules, how molecular speeds are statistically spread at a given T . Boltzmann provided the deeper framework: he generalized this to any energy form and any system, establishing that the probability of a microstate with energy E scales as $e^{(-E/kT)}$. The quantity kT is the characteristic thermal energy scale, when $\Delta E \gg kT$, higher states are exponentially depopulated; when $\Delta E \ll kT$, thermal fluctuations easily populate both states. kT is essentially the currency of all of thermal physics.

Q2. The lecture derives the Boltzmann distribution starting from a column of gas in a gravitational field. Explain the physical logic of this derivation: why does the number density of gas molecules decrease with height, and what mathematical form does this give for $n(h)$?

Ans: At the base of a gas column, gravitational potential energy $E(h) = mgh$ is low, so molecular density is high. As height increases, $E(h)$ increases. Balancing the upward pressure gradient ($dp/dh = dn/dh \cdot kT$) against the downward gravitational force ($-mgn$) gives $dn/dh = -(mg/kT)n$ an exponential ODE whose solution is $n(h) \propto \exp(-mgh/kT)$. This directly demonstrates the Boltzmann principle: population is exponentially suppressed by energy relative to thermal energy kT .

Q3. Distinguish between the Verlet and Velocity Verlet integration algorithms in terms of accuracy, storage, and availability of velocities.

Ans: Both are equivalent second-order integrators with $O(\Delta t^4)$ local truncation error. The original Verlet algorithm calculates positions using $r(t+\Delta t) = 2r(t) - r(t-\Delta t) + a(t)\Delta t^2$, it requires storing two position arrays and does not directly provide velocities (they must be estimated numerically, reducing velocity accuracy). Velocity Verlet simultaneously integrates positions and velocities in a single step, storing only one set of positions and one set of velocities — making it more memory-efficient and providing velocities at the same time level as positions, which is important for accurate kinetic energy and temperature calculation.

Q4. In PBC, the system is surrounded by infinite periodic images in all directions, but only the nearest image interactions are computed.

Ans: In Periodic Boundary Conditions (PBC), the simulation box is replicated infinitely in all directions to mimic a bulk system. However, to avoid double counting and reduce computational cost, interactions are calculated using the minimum image convention, where each particle interacts only with the closest periodic image of every other particle.

Q5. The scoring function in AutoDock Vina reports binding affinity as a negative value in kcal/mol (e.g., -7.82 kcal/mol). Connect this to the concept of Gibbs free energy and the Boltzmann distribution discussed in the Biophysics course.

Ans: The binding affinity reported by AutoDock Vina (e.g., -7.82 kcal/mol) is an estimate of the change in Gibbs free energy (ΔG) upon ligand binding. A more negative binding affinity corresponds to a more negative Gibbs free energy (ΔG), indicating a spontaneous and more stable ligand–receptor complex. According to the Boltzmann distribution ($P \propto e^{-E/kT}$), lower energy states are more populated, so a more negative ΔG means a higher proportion of molecules exist in the bound state compared to the unbound state.